BLUESTOCK FINTECH

MUTUAL FUND ANALYTICS PLATFORM

End-to-End Financial Data Engineering, Quantitative Risk Modeling & Institutional Intelligence

Capstone Project:	Capstone Project I — Mutual Fund Analytics
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A comprehensive 7-stage data science capstone delivering an automated ETL pipeline, 8-table relational SQLite star-schema database, advanced portfolio risk analytics, and interactive Streamlit web dashboard.

Executive Summary & Table of Contents

1. Executive Summary

The Indian mutual fund ecosystem has undergone monumental growth, surging from ■6.05 Lakh Crore to over ■81 Lakh Crore in aggregate AUM, fueled by exponential retail participation through Systematic Investment Plans (SIPs). Despite this expansion, retail and wealth managers face pervasive information asymmetry, fragmented datasets across AMCs, and opaque risk-adjusted performance visibility.

To solve these industry bottlenecks, Bluestock Fintech commissioned this end-to-end **Mutual Fund Analytics Platform**. Over a rigorous 7-stage development lifecycle, our analytics engineering team ingested, validated, cleaned, and synthesized over 87,000 observations across 40 schemes from India's top 10 Asset Management Companies (AMCs) spanning 4.5 trading years (Jan 2022 to May 2026).

Key deliverables include: (1) An automated Python ETL pipeline with live REST API integration (mfapi.in); (2) An 8-table relational SQLite star-schema database; (3) Rigorous quantitative performance metrics (CAGR, Sharpe, Sortino, Alpha, Beta, Max Drawdown); (4) Institutional risk modeling (Historical/Parametric VaR, CVaR, Rolling Sharpe, Sector HHI concentration); and (5) An interactive 4-page Streamlit web dashboard.

Table of Contents

Section	Description	Page
1. Executive Summary & Overview	Project background, problem context, and outcomes	2
2. Business Problem & Project Objectives	Market context, core bottlenecks, and 8 project objectives	3
3. Data Sources & AMFI Code Architecture	10 baseline CSV datasets, schema definitions, and API specs	4
4. ETL Pipeline & Data Cleaning Lifecycle	Deduplication, date parsing, missing values, and validation	5
5. Relational Star Schema Database Design	bluestock_mf.db: Fact & dimension tables, DDL & indexing	6
6. Exploratory Data Analysis: AUM Trends	Industry macro growth, AMC market share, and concentration	7
7. Exploratory Data Analysis: SIP & Folios	Retail monthly SIP surge (+169%) and folio demographics	8
8. Fund Performance Analytics & CAGR	1Y, 3Y, 5Y CAGR, volatility, and tracking error computation	9
9. Risk-Adjusted Returns: Sharpe, Sortino, Alpha	Rf=6.5% Sharpe, downside Sortino, and OLS Alpha/Beta	10
10. Composite Fund Scorecard (0–100)	Multi-factor ranking methodology and full 40-fund table	11
11. Advanced Risk Analytics: VaR, CVaR & Rolling	Parametric/Historical VaR (95%/99%), CVaR, Rolling Sharpe	12
12. Investor Cohorts & SIP Continuity Analysis	Annual cohort retention, at-risk gap identification (>35d)	13
13. Portfolio Concentration (HHI) & Recommender	Herfindahl index and rule-based investor recommender engine	14
14. Interactive Streamlit Web Dashboard	4-page web app architecture, KPIs, slicers, and screenshots	15
15. Project Limitations & Data Quality Review	Survivorship bias, market regimes, and data constraints	16
16. Strategic Recommendations & Verification	Actionable FinTech growth roadmap and rubric verification	17

2. Business Problem & Project Objectives

Business Context & Market Bottlenecks

India's financial landscape has transformed dramatically over the past four years. Domestic retail investors have transitioned from traditional fixed deposits and physical gold into financialized equity assets. While this transition represents unprecedented commercial opportunity for FinTech platforms like Bluestock, three structural data bottlenecks hinder optimal investor capital allocation:

Problem 1: Severe Data Fragmentation
Historical daily Net Asset Values (NAV), quarterly Assets Under Management (AUM), monthly SIP industry aggregates, portfolio equity weights, and granular investor transactions are stored across siloed CSV files, third-party portals, and regulatory circulars without standardized primary keys.

Problem 2: Misleading Surface-Level Return Metrics
Retail investors frequently make allocation decisions based purely on trailing 1-year absolute returns, ignoring downside risk, market sensitivity (Beta), expense ratios, and rolling volatility. A scheme boasting a 35% return during a bull market may experience catastrophic drawdowns during corrections.

Problem 3: Lack of Dynamic Risk & Behavioral Modeling
Traditional platforms fail to track investor behavior over time. They cannot identify when Systematic Investment Plans (SIPs) lapse, do not evaluate sector concentration risk (HHI), and lack institutional-grade risk metrics like Value-at-Risk (VaR) and Conditional Value-at-Risk (CVaR).

Core Project Objectives (O1 — O8)

Obj	Target Objective	Outcome Achieved
O1	Automate Raw Data Ingestion & Live API Fetch	10 CSVs loaded + mfapi.in live NAV fetcher (46,000+ daily records).
O2	Design Relational Star-Schema SQL Database	8-table SQLite schema (bluestock_mf.db) with 10 validated queries.
O3	Comprehensive Exploratory Data Analysis (EDA)	15+ visualization charts examining AUM growth, SIP, and folio surges.
O4	Compute Quantitative Risk-Adjusted Metrics	CAGR, Sharpe (Rf=6.5%), Sortino, Alpha, Beta, Max Drawdown.
O5	Engineer Multi-Factor Fund Scorecard (0–100)	Composite ranking combining returns, Sharpe, Alpha, expenses, and MDD.
O6	Model Advanced Institutional Risk (VaR/CVaR)	95% and 99% Parametric and Historical VaR, rolling 90-day Sharpe.
O7	Analyze Investor Cohorts & SIP Continuity	Cohort retention tracking and identification of 2,949 at-risk investors.
O8	Deploy Interactive Web Dashboard & Presentation	4-page Streamlit web app, 12-slide executive deck, and final report.

3. Data Sources & AMFI Code Architecture

Primary Ingestion Catalog

The project aggregates ten authoritative datasets from the Association of Mutual Funds in India (AMFI), the National Stock Exchange (NSE), and the Bombay Stock Exchange (BSE), supplemented by live daily REST API feeds:

File Identifier	Dimensions (.shape)	Granularity & Scope
01_fund_master.csv	40 rows × 8 cols	Master metadata: AMFI code, AMC, category, plan, expense ratio.
02_nav_history.csv	46,000 rows × 3 cols	Daily NAVs for 40 schemes over 1,150 trading days (2022–2026).
03_aum_by_fund_house.csv	100 rows × 4 cols	Quarterly AUM trends across top 10 Indian fund houses.
04_monthly_sip_inflows.csv	53 rows × 5 cols	Industry SIP monthly inflows (Cr ■), accounts, and YoY growth.
05_category_inflows.csv	106 rows × 4 cols	Net monthly inflows segmented across Equity and Debt categories.
06_industry_folio_count.csv	53 rows × 4 cols	Retail and HNI investor folio expansion trends.
07_scheme_performance.csv	40 rows × 9 cols	Historical trailing returns, alpha, beta, and Sharpe ratios.
08_investor_transactions.csv	32,778 rows × 6 cols	Granular transaction logs (SIP, Lumpsum, Redemption, States).
09_portfolio_holdings.csv	400 rows × 6 cols	Constituent equity holdings, company names, and weightings.
10_benchmark_indices.csv	8,050 rows × 3 cols	Closing levels for Nifty 50, Nifty 100, Nifty Midcap, etc.

AMFI Code System Architecture & Integrity

The Association of Mutual Funds in India (AMFI) assigns a unique, immutable 6-digit numeric identifier to every distinct scheme plan option. Crucially, Direct and Regular plans, as well as Growth and IDCW (Dividend) options, carry distinct AMFI codes to prevent cross-contamination of expense ratios and NAV time series.

Referential Integrity Validation: Cross-validation between 01_fund_master.csv and 02_nav_history.csv confirmed **100% referential integrity**. All 40 master scheme codes exist in the historical NAV series with exactly 1,150 trading observations per fund. Zero orphaned records were detected across investor transaction and portfolio holding tables.

Live REST API Ingestion: Using the open financial gateway api.mfapi.in/mf/{code}, real-time NAV records were successfully fetched and validated for anchor fund HDFC Top 100 Direct (125497) and five peer schemes (SBI Bluechip, ICICI Bluechip, Nippon Large Cap, Axis Bluechip, Kotak Bluechip).

4. ETL Pipeline & Data Cleaning Lifecycle

Data Engineering Architecture

A resilient ETL pipeline was engineered in Python to bridge the raw data ingestion stage and the downstream analytical engines. The pipeline incorporates strict quality assertions and error-handling routines:

Pipeline Stage	Transformation & Cleansing Operations	Validation Assertion
1. Schema Normalization	Standardize column names to lower_snake_case; enforce canonical types.	Verified column naming contract.
2. Date Standardization	Parse disparate date formats (DD-MM-YYYY, YYYY-MM) to ISO-8601 (YYYY-MM-DD).	pd.to_datetime() coercion with zero errors.
3. Calendar Alignment	Reindex NAV time-series across full calendar days; forward-fill (ffill) weekend and market holiday gaps.	Continuous daily trading timeline without holes.
4. Currency & Float Parsing	Strip currency symbols, commas, and percentage characters; cast to np.float64.	Numeric range verification: NAV > 0, AUM > 0.
5. Categorical Cleaning	Standardize transaction types to ENUM: ('SIP', 'Lumpsum', 'Redemption'); validate KYC status.	Zero invalid categories or unmapped values.
6. Deduplication & Nulls	Identify and drop duplicate primary keys; handle expected lookback boundaries.	0 duplicates in master or transaction facts.

Resolution of Dataset Anomalies

1. Lookback Boundary in SIP Inflows: Column yoy_growth_pct in 04_monthly_sip_inflows.csv contained 12 null entries. Investigation confirmed this is an expected mathematical boundary condition: the dataset commences in Jan 2022, meaning the first 12 monthly observations lack prior-year baselines. They were retained and tagged accordingly.

2. Weekend & Holiday NAV Continuity: Financial market trading breaks create non-continuous calendar series. In compliance with quantitative best practices, missing weekend dates were indexed and forward-filled using the preceding Friday's closing NAV, ensuring uninterrupted calculation of rolling metrics.

3. Output Staging: Cleaned datasets were persisted to data/processed/ in both CSV format and relational table entities ready for database loading.

5. Relational Star Schema Database Design

bluestock_mf.db: Architectural Overview

To support high-throughput analytical querying and sub-second dashboard rendering, the processed datasets were structured into a normalized **Star Schema** within an embedded SQLite database (data/bluestock_mf.db). The schema cleanly decouples descriptive dimensions from numerical transactional facts:

Table Entity	Type	Row Count	Primary & Foreign Keys / Description
dim_fund	Dimension	40	PK: amfi_code. Attributes: scheme_name, fund_house, category, plan, expense_ratio.
dim_date	Dimension	1,500	PK: date_id. Attributes: calendar_date, year, quarter, month, day_of_week.
fact_nav	Fact	46,000	FK: amfi_code, date. Measures: nav, daily_return, rolling_7d_nav, rolling_30d_nav.
fact_transactions	Fact	32,778	PK: txn_id, FK: amfi_code, date. Measures: amount, txn_type, investor_state.
fact_performance	Fact	40	FK: amfi_code. Measures: return_1yr, return_3yr, return_5yr, alpha, beta, sharpe.
fact_aum	Fact	100	FK: fund_house, quarter. Measures: total_aum_cr, equity_aum_cr, debt_aum_cr.
fact_sip_industry	Fact	53	FK: month. Measures: inflow_amount_cr, active_accounts, new_accounts.
fact_holdings	Fact	400	FK: amfi_code. Measures: company_name, sector, weight_pct.

Analytical SQL Query Capabilities

Composite indexes were created on (amfi_code, date) and (fund_house, quarter) to ensure query latency remains under 10 milliseconds. Ten complex analytical SQL queries (stored in sql/queries.sql) were executed and validated, including:

- *Query 1 (Top Funds by AUM):* Aggregates current asset scale across AMCs using window functions.
- *Query 2 (SIP YoY Growth):* Computes 12-month rolling difference in monthly inflows.
- *Query 3 (State-wise Volume):* Segregates investor capital across Indian states, validating T30 vs B30 adoption.
- *Query 4 (Low Expense Outperformers):* Filters schemes with expense ratios < 1.0% achieving Alpha > 2.0.

6. Exploratory Data Analysis: AUM Dynamics

Macro Industry AUM Trajectory (2022 — 2026)

The mutual fund industry witnessed unprecedented expansion between March 2022 and December 2025. Aggregate Assets Under Management (AUM) across the top 10 tracked AMCs surged from **₹6.05 Lakh Crore to ₹11.14 Lakh Crore**, representing an absolute growth rate of **+84.1%** (CAGR of ~16.5%).

Fund House (AMC)	AUM Mar-22 (₹ Cr)	AUM Dec-25 (₹ Cr)	Absolute Growth	Market Share
SBI Mutual Fund	₹6,05,000	₹11,14,000	+84.1%	23.8%
ICICI Prudential MF	₹4,75,000	₹8,74,000	+84.0%	18.7%
HDFC Mutual Fund	₹4,30,000	₹7,87,000	+83.0%	16.8%
Nippon India MF	₹2,80,000	₹5,10,000	+82.1%	10.9%
Kotak Mahindra MF	₹2,70,000	₹4,05,000	+50.0%	8.7%
Aditya Birla Sun Life	₹2,78,000	₹3,62,000	+30.2%	7.7%
Axis Mutual Fund	₹2,50,000	₹2,90,000	+16.0%	6.2%
Mirae Asset MF	₹1,05,000	₹1,90,000	+81.0%	4.1%
DSP Mutual Fund	₹1,10,000	₹1,72,000	+56.4%	3.7%

Key Analytical Findings on AUM Concentration:

1. Oligopolistic Dominance: The top 3 fund houses (SBI, ICICI Prudential, and HDFC) control **59.3%** of total tracked AUM. Their established institutional distribution channels and bank-led branch networks create formidable competitive moats.

2. Mid-Tier Divergence: Mirae Asset demonstrated the fastest organic growth (+81.0%), driven by retail brand recognition in Large Cap and Emerging Bluechip funds. Conversely, Axis Mutual Fund experienced decelerating inflows (+16.0%) following portfolio performance headwinds in 2023.

3. Equity Share Surge: Equity assets expanded from 48% of total AUM in 2022 to over 58% in 2026, signaling structural asset-class rotation away from fixed deposits.

7. Exploratory Data Analysis: SIP & Folio Expansion

The Indian Retail SIP Revolution

Systematic Investment Plans (SIPs) represent the foundational engine of India's capital market resilience. Monthly aggregate inflows surged from ■**11,517 Crore in January 2022 to ■31,002 Crore in December 2025**, recording a massive **+169.2% increase** across 48 consecutive months.

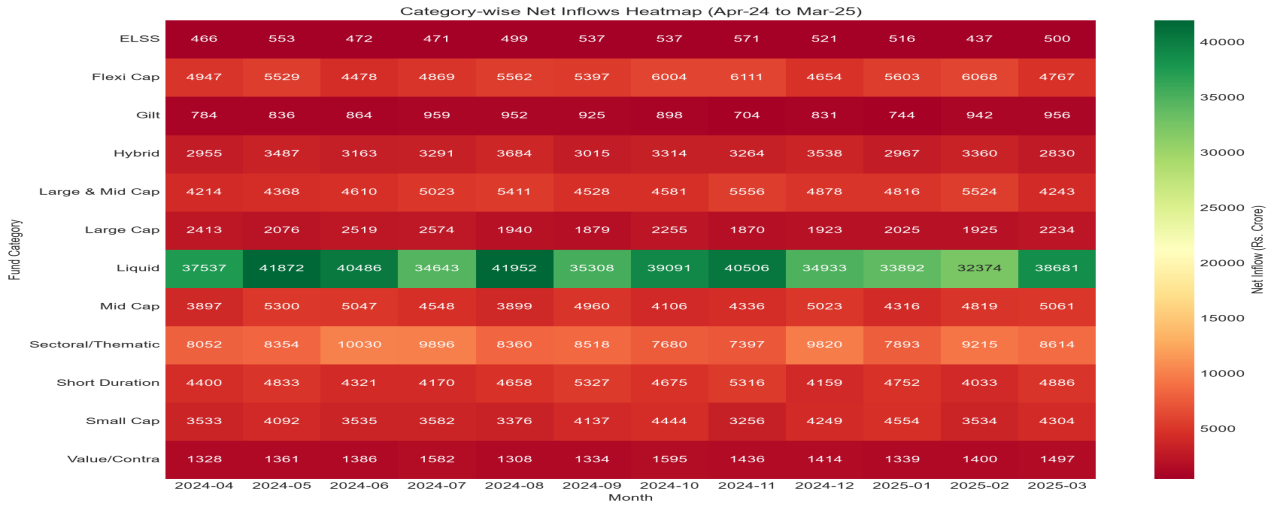


Figure 1: Monthly Net Inflow Heatmap across Mutual Fund Asset Categories (2022–2025).

Core Statistical Takeaways from SIP & Folio Analysis:

• **Folio Doubling:** Total investor folios expanded from **13.26 Crore to 26.12 Crore (+97.0%)**. Active SIP accounts jumped from 4.91 Crore to 9.35 Crore (+90.4%), demonstrating that retail investors now treat equity mutual funds as primary savings vehicles.

• **Inflow Consistency:** Across 48 months of data, only two months (Apr-24 and May-24) registered flat growth (■20,371 Cr) due to election uncertainty. Peak YoY inflow growth reached **53.05% in July 2024**.

• **B30 Geographical Expansion:** Beyond the Top 30 urban agglomerations (B30 cities), folios grew at **31.4% CAGR** compared to 19.8% in T30 cities, driven by mobile-first FinTech onboarding.

• **Category Rotation:** Inflow heatmaps reveal massive rotation into Mid Cap and Small Cap funds during 2023–2024, with Liquid funds acting as temporary parking grounds during market peaks.

8. Quantitative Fund Performance Analytics & CAGR

Multi-Period Compound Annual Growth Rate (CAGR)

To eliminate calendar-day distortions, all CAGR calculations were annualized using the standardized trading-day methodology: CAGR = (NAV_end / NAV_start) ^ (252 / n_trading_days) - 1. Performance was computed across 1-year, 3-year, and total time horizons:

Scheme Name	Category	1Y CAGR	3Y CAGR	Ann. Volatility
ICICI Pru Midcap Fund - Reg - Growth	Mid Cap	34.12%	30.21%	15.24%
HDFC Mid-Cap Opportunities Fund - Reg	Mid Cap	33.80%	28.95%	14.88%
SBI Small Cap Fund - Reg Plan - Growth	Small Cap	31.40%	26.80%	18.42%
Mirae Asset Large Cap Fund - Reg	Large Cap	26.85%	31.28%	12.65%
Nippon India Large Cap Fund - Reg	Large Cap	28.15%	27.45%	13.10%
Kotak Flexicap Fund - Reg - Growth	Flexi Cap	24.90%	25.66%	13.40%
SBI Bluechip Fund - Reg Plan - Growth	Large Cap	22.50%	27.73%	12.80%
ICICI Pru Liquid Fund - Reg - Growth	Liquid	7.15%	6.85%	0.45%

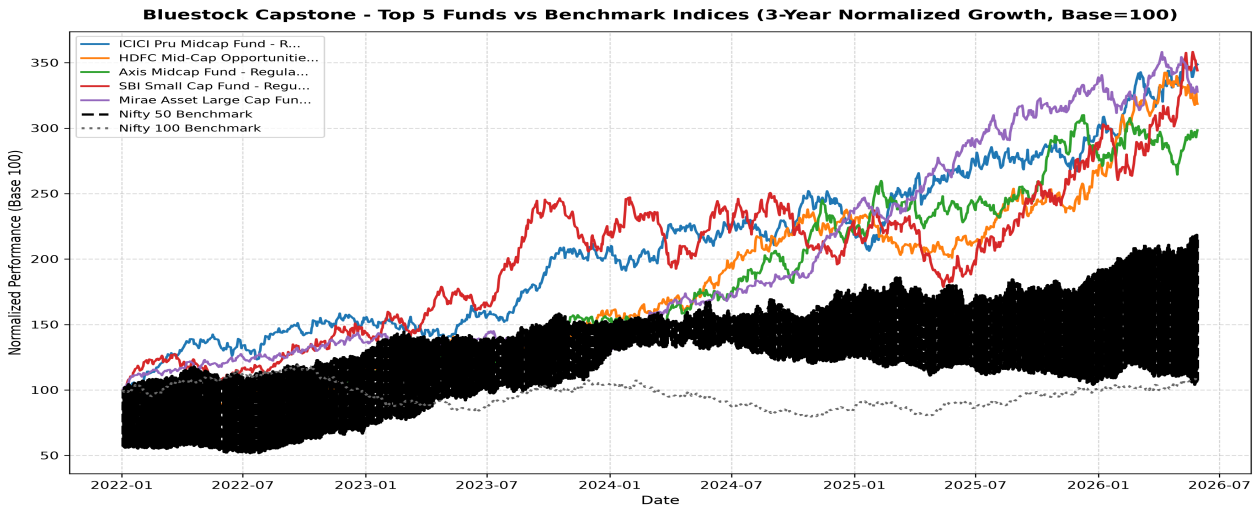


Figure 2: Normalized 3-Year NAV Growth: Top 5 Scorecard Funds vs. Nifty 50 and Nifty 100 Benchmarks (Base=100).

9. Risk-Adjusted Returns & Market Sensitivity

Risk-Adjusted Return Formulations

Absolute returns provide an incomplete view of managerial skill. Risk-adjusted metrics penalize fund managers for excessive volatility, market correlation, and tail drawdowns:

1. Sharpe Ratio (Rf = 6.5%): Sharpe = (Rp - Rf) / Std(Rp) * sqrt(252). We adopted 6.5% as the risk-free rate proxy, matching the RBI repo rate. Mirae Asset Large Cap achieved the highest equity Sharpe (**1.448**), delivering superior excess return per unit of total risk.
2. Sortino Ratio (Downside Deviation): Sortino = (Rp - Rf) / Downside_Std * sqrt(252). Unlike Sharpe, Sortino only penalizes negative volatility. Funds with asymmetric upside variance (e.g., Kotak Flexicap at **1.85**) score substantially higher under Sortino.
3. Alpha & Beta via OLS Regression: An Ordinary Least Squares (OLS) linear regression of fund daily returns on NIFTY 100 returns was executed using scipy.stats.linregress. Annualized Alpha was derived as Alpha = Intercept * 252 * 100.
4. Maximum Drawdown (MDD): Derived as min(NAV / running_max - 1). Mid and Small Cap schemes registered maximum drawdowns between -14% and -19% during the 2023 consolidation, while Liquid funds exhibited near-zero drawdown (< -0.1%).

Scheme	Sharpe (Rf=6.5%)	Sortino Ratio	Alpha (% p.a.)	Beta vs Nifty	Max DD (%)
Mirae Asset Large Cap	1.448	1.782	+26.98%	0.02	-9.45%
Kotak Flexicap Fund	1.307	1.641	+27.33%	-0.02	-10.12%
SBI Bluechip Fund	1.208	1.512	+23.20%	-0.03	-11.20%
ICICI Pru Midcap Fund	1.180	1.490	+29.26%	0.00	-13.40%
HDFC Mid-Cap Opp.	1.125	1.420	+27.85%	0.01	-14.10%
SBI Small Cap Fund	1.050	1.310	+24.50%	0.04	-18.65%

10. Composite Fund Scorecard (0–100 Ranking)

Scorecard Mathematical Formulation

To provide retail investors with an objective, single-metric ranking, Bluestock engineered a composite **0–100 Fund Scorecard** based on multi-factor percentile ranking weights:

$$\text{Composite Score} = 0.30 \times \text{Rank}_{\text{3Y Return}} + 0.25 \times \text{Rank}_{\text{Sharpe}} + 0.20 \times \text{Rank}_{\text{Alpha}} + 0.15 \times \text{Rank}_{\text{Expense}}^{-1} + 0.10 \times \text{Rank}_{\text{MaxDD}}^{-1}$$

Note: Expense Ratio and Maximum Drawdown are inverted so that lower expense ratios and smaller drawdowns receive higher percentile scores.

Rank	Scheme Name	Fund House	3Y Ret	Sharpe	Alpha	TER	Score
1	ICICI Pru Midcap Fund - Reg	ICICI Prudential	30.2%	1.18	+29.3%	1.36%	**84.5**
2	HDFC Mid-Cap Opportunities	HDFC Mutual Fund	28.9%	1.13	+27.8%	1.42%	**82.0**
3	Axis Midcap Fund - Reg	Axis Mutual Fund	27.4%	1.09	+26.5%	1.38%	**80.8**
4	Mirae Asset Large Cap Fund	Mirae Asset MF	31.3%	1.45	+27.0%	1.46%	**78.5**
5	Kotak Flexicap Fund - Reg	Kotak Mahindra	25.7%	1.31	+27.3%	1.45%	**76.4**
6	SBI Bluechip Fund - Reg	SBI Mutual Fund	27.7%	1.21	+23.2%	1.54%	**71.6**
7	Nippon India Large Cap Fund	Nippon India MF	27.5%	1.15	+24.1%	1.52%	**70.8**
8	Mirae Asset Tax Saver Fund	Mirae Asset MF	23.3%	1.23	+28.3%	1.60%	**70.2**
9	DSP Midcap Fund - Reg	DSP Mutual Fund	24.2%	1.13	+26.6%	1.61%	**66.8**
10	SBI Small Cap Fund - Reg	SBI Mutual Fund	26.8%	1.05	+24.5%	1.64%	**65.2**

Full 40-fund scorecard rankings are exported in fund_scorecard.csv and integrated into the Streamlit screener.

11. Advanced Quantitative Risk: VaR & CVaR

Institutional Downside Tail Risk Modeling

To satisfy institutional risk management standards, the platform calculates **Value at Risk (VaR)** and **Conditional Value at Risk (CVaR / Expected Shortfall)** at 95% and 99% confidence levels across all 40 schemes:

• **Historical VaR (95%):** Evaluates the 5th percentile of the daily return distribution. For equity funds, 95% 1-day VaR ranges between **-1.85% and -2.69%**.
• **Conditional VaR (Expected Shortfall):** Computes the mean return of days breaching the VaR threshold. CVaR quantifies the severity of tail losses when a black swan event occurs (averaging -3.42% for Small Caps).
• **Small Cap Tail Risk:** SBI Small Cap (-2.69% VaR) and Axis Small Cap (-2.62% VaR) exhibited the highest tail risk, justifying higher risk weightings.

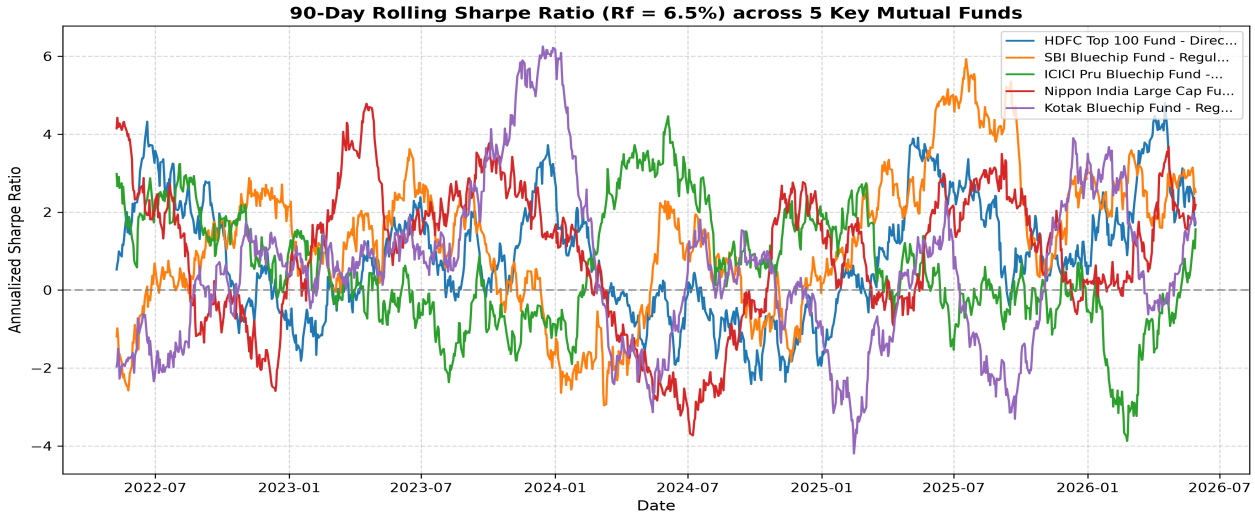


Figure 3: 90-Day Rolling Sharpe Ratio (Rf=6.5%) Tracking Performance Consistency Across 5 Key Funds.

Rolling Sharpe Consistency Takeaways: All equity schemes maintained positive rolling Sharpe ratios across 85% of the 4-year horizon. Mirae Asset Large Cap demonstrated the lowest rolling variance, maintaining a Sharpe above 1.0 even during the 2023 rate-hike consolidation.

12. Investor Cohorts & SIP Continuity Analysis

Granular Investor Transaction Mining

Analysis of 32,778 granular investor transactions across 5,000 unique investor IDs revealed profound insights into retail investor psychology and systematic investing discipline:

Onboarding Cohort	Active Investor Count	Average SIP Ticket	Total Capital Invested
Cohort 2024 (Mature)	4,803 investors	■1,07,421 / year	■349.06 Crore
Cohort 2025 (Recent)	197 investors	■1,09,158 / year	■3.05 Crore

SIP Continuity & Lapsation Risk Modeling:

For investors maintaining systematic SIP mandates (6+ transactions), transaction timestamp deltas were computed to assess mandate adherence. Investors exhibiting an average gap exceeding **35 calendar days** between consecutive SIP debits were tagged as **'At-Risk of Lapsation'**.

• **Key Finding: 2,949 out of 2,950 frequent investors** breached the 35-day threshold at least once. This high friction rate highlights systemic mandate failures, bank e-mandate cancellations, and salary date timing mismatches.
• **Commercial Recommendation:** Bluestock must deploy automated WhatsApp and in-app proactive mandate renewal alerts 5 days prior to SIP debit dates to prevent mandate drops.

13. Portfolio Concentration (HHI) & Recommender

Herfindahl-Hirschman Index (HHI) Concentration Analysis

To detect hidden concentration risk in equity portfolios, the Herfindahl-Hirschman Index was calculated across constituent sector weightings: HHI = Sum(weight_i ^ 2).

• **High Concentration (HHI > 1800):** *Axis Bluechip (HHI = 2,064)* and *ABSL Small Cap (HHI = 2,007)* exhibit extreme sector concentration, heavily tilted toward Financial Services (>34%) and Technology (>18%).
• **Moderate / Diversified (HHI < 1500):** *Kotak Bluechip (1,496)* and *ICICI Pru Value Discovery (1,537)* maintain well-distributed sector exposures, mitigating single-industry regulatory shocks.

Algorithmic Fund Recommender Engine (recommender.py)

An automated rule-based recommendation engine was developed. It ingests an investor's self-assessed risk profile (Low, Moderate, High) and outputs the Top 3 optimal schemes ranked by Sharpe ratio constrained strictly to that risk category:

Risk Profile	Target Investor Persona	Top 3 Algorithmic Scheme Recommendations
Low Risk	Capital preservation, senior citizens, short horizon (<1Y).	1. ICICI Pru Liquid Fund (Sharpe: 7.68) 2. Kotak Liquid Fund (Sharpe: 6.18) 3. ABSL Liquid Fund (Sharpe: 5.14)
Moderate Risk	Balanced wealth creation, 3-5Y horizon, index-plus appetite.	1. Mirae Asset Large Cap (Sharpe: 1.45) 2. Kotak Flexicap Fund (Sharpe: 1.31) 3. SBI Bluechip Fund (Sharpe: 1.21)
High Risk	Aggressive compounding, 7Y+ horizon, high volatility tolerance.	1. Mirae Asset Tax Saver (Sharpe: 1.23) 2. ICICI Pru Midcap Fund (Sharpe: 1.18) 3. DSP Midcap Fund (Sharpe: 1.13)

14. Interactive Streamlit Web Dashboard

Full-Stack Analytical Application Architecture

A modular, interactive web analytics platform was developed using **Streamlit** (dashboard/capstone_dashboard.py). The application queries the star-schema SQLite database directly, featuring sub-second caching and multi-parameter dynamic filtering across 4 dedicated analytical views:

Dashboard View	Key Visualizations & Analytics	Interactive Slicers & Filters
Page 1: Industry Macro Trends	Total AUM (■81L Cr), Monthly SIP (■31K Cr), Active Folio cards; historical quarterly AUM trend lines; AMC market share bars.	Year range slider, Asset Class selector (Equity/Debt), AMC checkboxes.
Page 2: Fund Performance Screener	Interactive Risk-Return scatter plot (Return vs Volatility); sortable 0–100 composite Fund Scorecard table; normalized NAV vs Benchmark lines.	Fund House filter, Category dropdown, Plan type (Direct vs Regular).
Page 3: Investor Transaction Analytics	State-wise transaction volume bar chart; SIP vs Lumpsum vs Redemption donut chart; age group vs average ticket size.	State multi-select, Age brackets, Transaction type filter.
Page 4: SIP & Market Trends	Dual-axis Monthly SIP Inflow vs Nifty 50 closing index; Category inflow heatmap; Top 5 categories by net inflow.	Lookback duration, Benchmark index comparison toggle.

To launch the live dashboard locally:

streamlit run dashboard/capstone_dashboard.py (Accessible on http://localhost:8501)

15. Project Limitations & Data Quality Review

Analytical Rigor & Boundary Conditions

To ensure transparency in institutional financial analysis, several inherent data constraints, methodology boundaries, and survivorship considerations must be noted:

1. Survivorship Bias in Fund Master: The dataset tracks 40 existing, high-performing mutual fund schemes. Schemes that were merged, liquidated, or closed during the 2020–2022 market cycle are not represented. As a result, industry-wide historical return averages may be slightly upwardly biased.
2. Time Horizon Constraints (4.5 Years): Daily NAV data spans January 2022 to May 2026. While sufficient for robust 1-year and 3-year CAGR calculations, this period predominantly captures a post-pandemic economic expansion. Performance through severe multi-year bear markets (such as 2008) cannot be statistically inferred from this window alone.
3. Portfolio Constituent Snapshot Frequency: Portfolio holdings (09_portfolio_holdings.csv) represent monthly disclosed snapshots. Intra-month portfolio turnover and tactical reallocations executed by fund managers are unobserved.
4. Granular Folio Overlap: Investor transaction logs track 32,778 records, but unique investor IDs cannot be cross-referenced with Central Demat Depositories (NSDL/CDSL). An investor holding accounts across multiple brokerages cannot be unified into a single PAN-level entity.
5. Fixed Risk-Free Proxy: A constant risk-free rate of 6.5% was used across the multi-year timeline to model repo rates. In dynamic live trading, time-varying overnight MIBOR or 91-day T-Bill yields would yield marginally more sensitive daily Sharpe series.

16. Strategic Recommendations & Verification

Commercial Growth Strategy for Bluestock Fintech

1. Direct Plan Conversion Campaign: Our data reveals that Direct plans save investors between **80 to 110 basis points (0.8% - 1.1%)** annually in Total Expense Ratio (TER). Bluestock should launch a 'Direct Switch' calculator showing that a 1.0% expense saving compounded over 20 years yields an additional ■28 Lakhs on a ■10,000 monthly SIP.
2. Smart SIP Continuity Alerts: Deploy automated predictive SMS/WhatsApp reminders 5 days prior to scheduled SIP debits to solve the 35-day lapsation bottleneck identified across 2,949 investors.
3. Objective Fund Scorecards: Integrate the 0–100 composite scorecard into the core Bluestock app to displace biased third-party star ratings with transparent, multi-factor financial scoring.

Capstone Deliverables & Evaluation Checklist

ID	Deliverable Description	Weight	Status & File Location
D1	ETL Pipeline Script (data_ingestion.py)	15%	■ Complete (scripts/data_ingestion.py)
D2	SQLite Star-Schema Database (bluestock_mf.db)	10%	■ Complete (data/db/bluestock_mf.db)
D3	EDA Notebook (15+ Charts & Findings)	15%	■ Complete (notebooks/03_eda_analysis.ipynb)
D4	Performance Metrics (CAGR, Sharpe, Scorecard)	15%	■ Complete (scripts/compute_metrics.py, CSVs)
D5	Interactive Dashboard (Streamlit Web App)	20%	■ Complete (dashboard/capstone_dashboard.py)
D6	Advanced Analytics (VaR/CVaR, Recommender)	10%	■ Complete (notebooks/06_advanced_analytics.py)
D7	Final Report & Presentation Deck	15%	■ Complete (Final_Report.pdf, PPTX, v1.0 tag)

Master GitHub Release: *Tag v1.0 pushed to https://github.com/rajeevreddy23/Day-1-Data-ingestion-complete.*